

Project: Data Lake Implementation using AWS Glue and Athena

Services Used

Amazon S3,
AWS Glue,
Amazon Athena,
AWS Identity and Access Management

Description

This project demonstrates the implementation of a **serverless data lake architecture on AWS**. Raw data files are stored in Amazon S3, cataloged using AWS Glue, and queried using Amazon Athena without provisioning any database servers. The project also includes IAM configuration for secure access control.

Definition

A data lake is a centralized repository that allows storing structured and unstructured data at any scale. It enables data processing and analytics directly on stored data without requiring traditional database systems.

Scope of the Project

- Store multiple datasets in Amazon S3
- Organize data using folder-based structure
- Use AWS Glue crawler to automatically detect schema
- Create a centralized data catalog
- Query data using Amazon Athena with SQL
- Configure IAM roles and permissions
- Combine multiple files into a single logical dataset

This Project Covers

- Data lake architecture fundamentals
- S3 bucket and folder organization
- Glue crawler configuration
- Schema detection and cataloging
- Serverless SQL querying using Athena
- Handling multiple files as a single dataset
- IAM role-based access control

Methodology

1. Serverless Data Lake Architecture (Implemented Method)

- Uses S3 for storage
- Uses Glue for metadata and schema detection
- Uses Athena for querying
- No database servers required
- Fully serverless and scalable

2. Traditional Database Approach (Alternative Method)

- Data is stored inside RDS/MySQL
- Requires provisioning database servers
- Schema must be defined manually
- Less flexible for large-scale analytics

❑ Not used in this project because it is not scalable for big data use cases

AWS Services Used in the Project

1. Amazon S3

Definition: Object storage service for storing data.

Purpose: To act as the data lake storage layer.

Role: Stores CSV files in structured folders.

2. AWS Glue

Definition: Data integration service for cataloging data.

Purpose: To detect schema and create metadata tables.

Role: Converts raw files into structured tables using crawler.

3. Amazon Athena

Definition: Serverless query service using SQL.

Purpose: To query data directly from S3.

Role: Executes SQL queries using Glue Data Catalog.

4. AWS IAM

Definition: Access control service for AWS resources.

Purpose: To manage permissions securely.

Role: Controls access to S3, Glue, and Athena.

Project Data Details

Dataset Type: csv

sales1.csv

Data Structure

id,name,product,amount

1,John,Laptop,800

2,Alice,Phone,500

sales2.csv

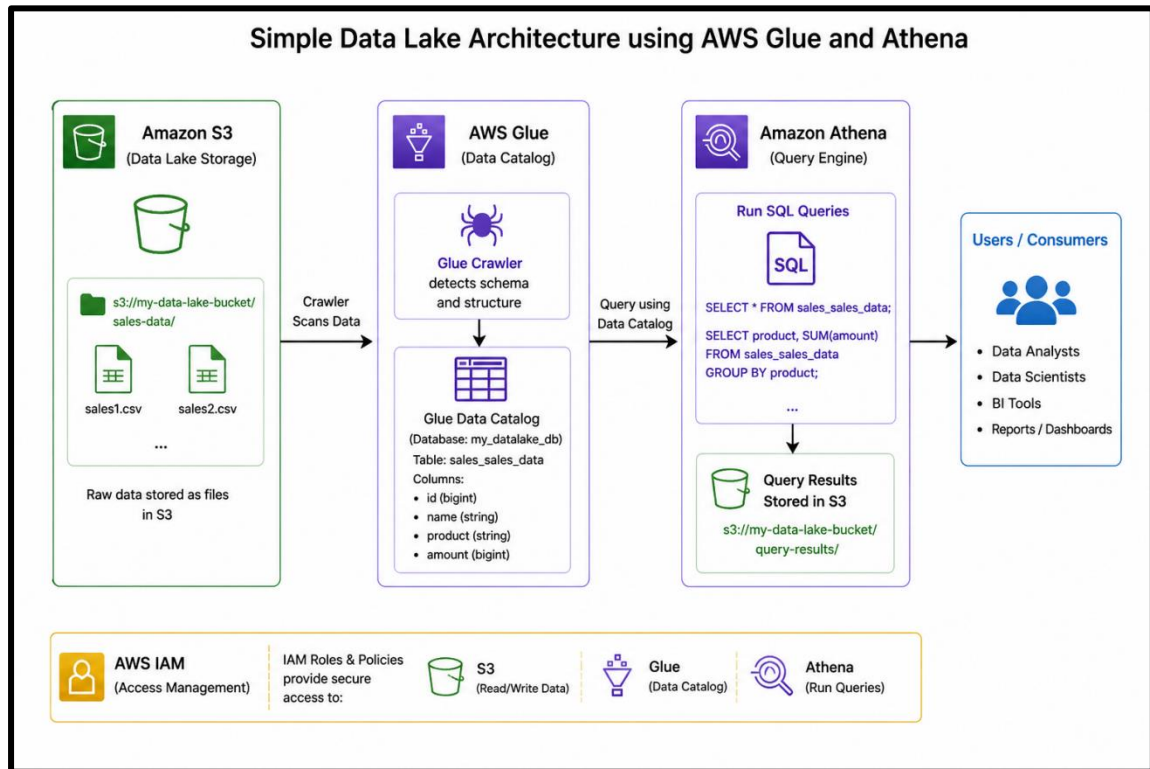
Data Structure

id,name,product,amount

3,Bob,Tablet,300

4,Eve,Laptop,1200

Data Lake Architecture:



Implementation Steps:

Create Role:

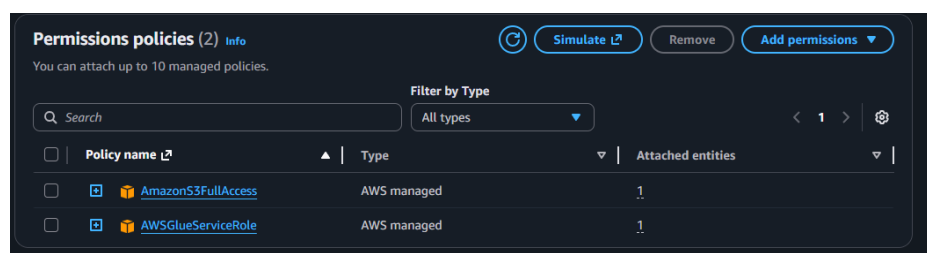
Steps:

1. IAM → Roles → Create Role
2. Trusted entity:
 1. AWS service
 2. Use case: **Glue**

Attach Policies:

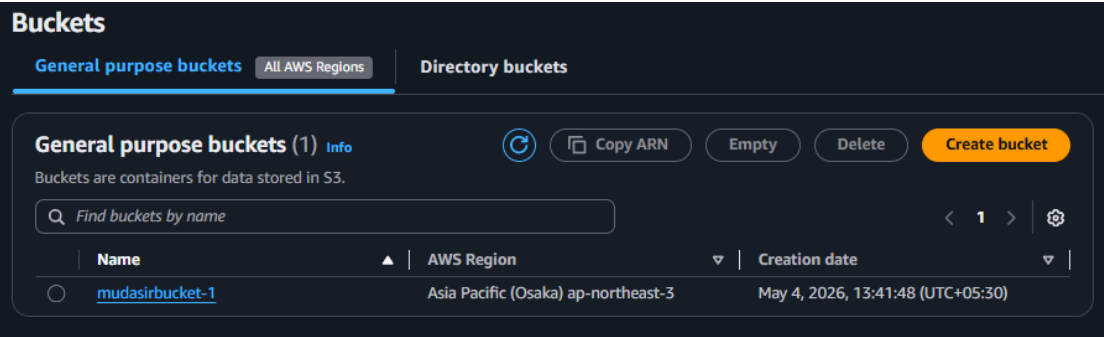
Required:

- AWSGlueServiceRole (mandatory)
- AmazonS3FullAccess (for learning)



Step 1: Create S3 Bucket

- Created a bucket:
- Mudasirbucket-1

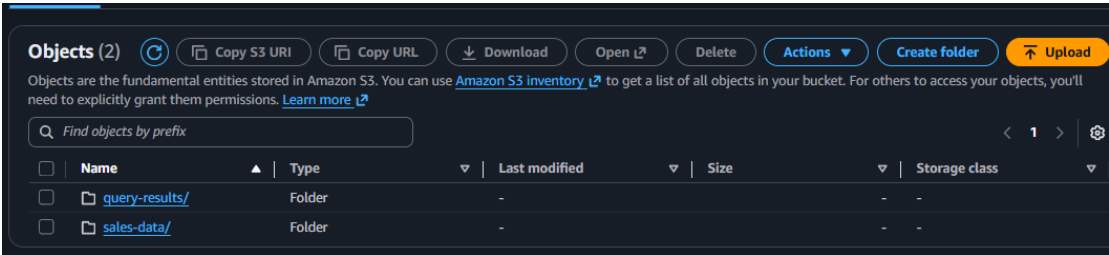


Purpose: To store raw data files.

Step 2: Create Folder Structure

Created folder:

`sales-data/`

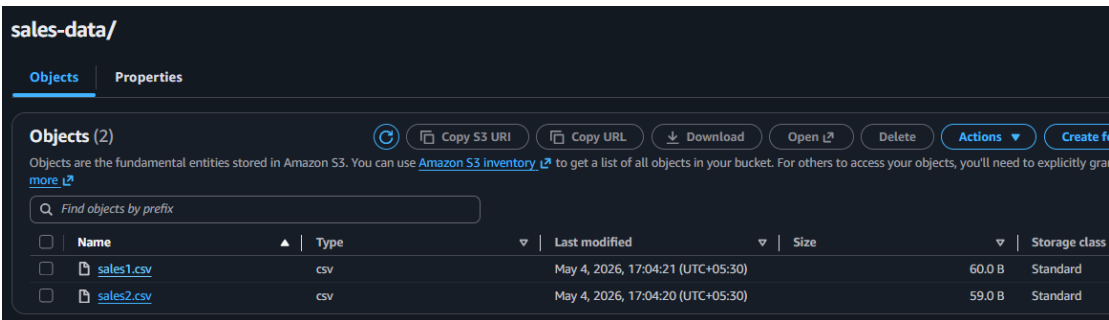


Purpose: To organize multiple files into a dataset.

Step 3: Upload Multiple Files

Uploaded:

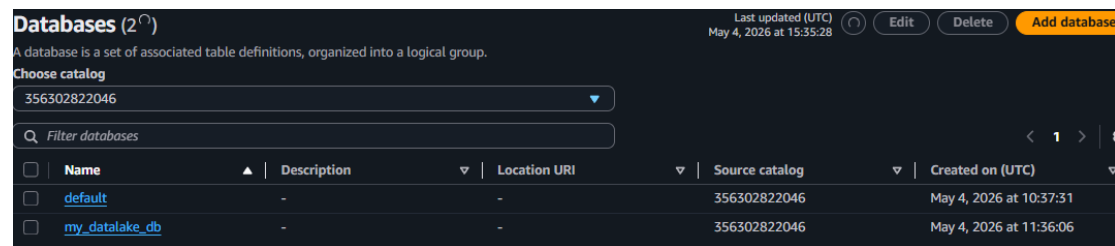
- sales1.csv
- sales2.csv



Purpose: To simulate real-world data lake with multiple files.

Step 4: Create Glue Database

- Created database:
- my_datalake_db

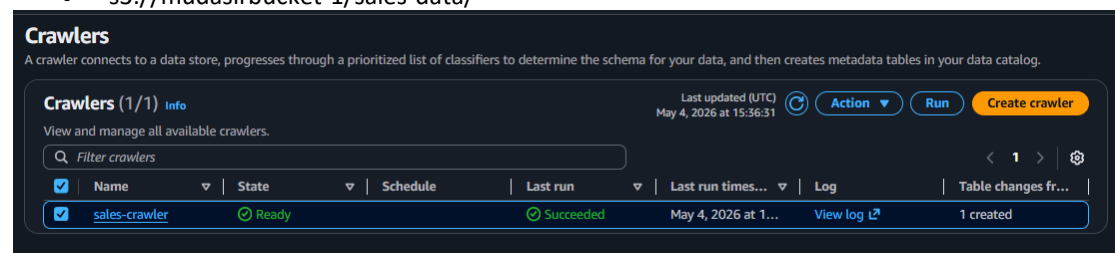


Purpose: To store metadata tables.

Step 5: Create Glue Crawler

Configured crawler with:

- Data source: S3
- Path:
- s3://mudasirbucket-1/sales-data/

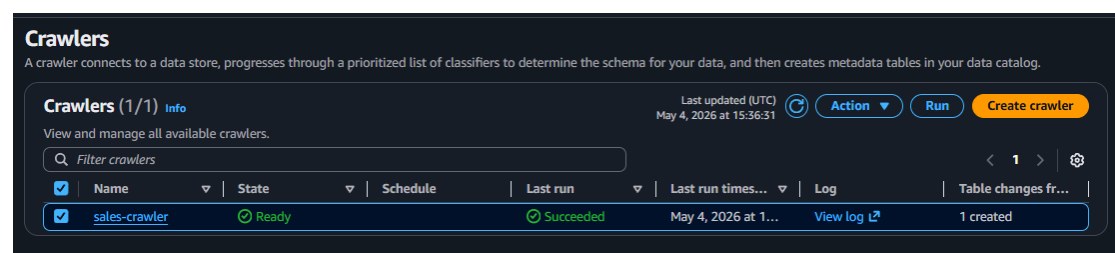


- IAM Role: glue-role

Purpose: To scan S3 data and detect schema automatically.

Step 6: Run Glue Crawler

- Executed crawler



Purpose: To generate metadata and create table.

Step 7: Verify Table in Glue

- Table created with columns:
 - id (bigint)
 - name (string)

- product (string)
- amount (bigint)

Purpose: To confirm schema detection.

#	Column name	Data type	Partition key	Comment
1	id	bigint	-	-
2	name	string	-	-
3	product	string	-	-
4	amount	bigint	-	-

Name	Database	Location	Classification	Deprecated
sales_data	my_datalake_db	s3://mudasirbucket	CSV	-

Step 8: Configure Athena Query Location

- Set query result location:
- s3://mudasirbucket-1/query-results/

Purpose: Athena stores query results in S3.

Location of query result - optional
Enter an S3 prefix in the current region where the query result will be saved as an object.

[View](#) [Browse S3](#)

You can create and manage lifecycle rules for this bucket
Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. [Learn more](#)

[Lifecycle configuration](#)

Step 9: Query Data in Athena

Executed queries:

View all data

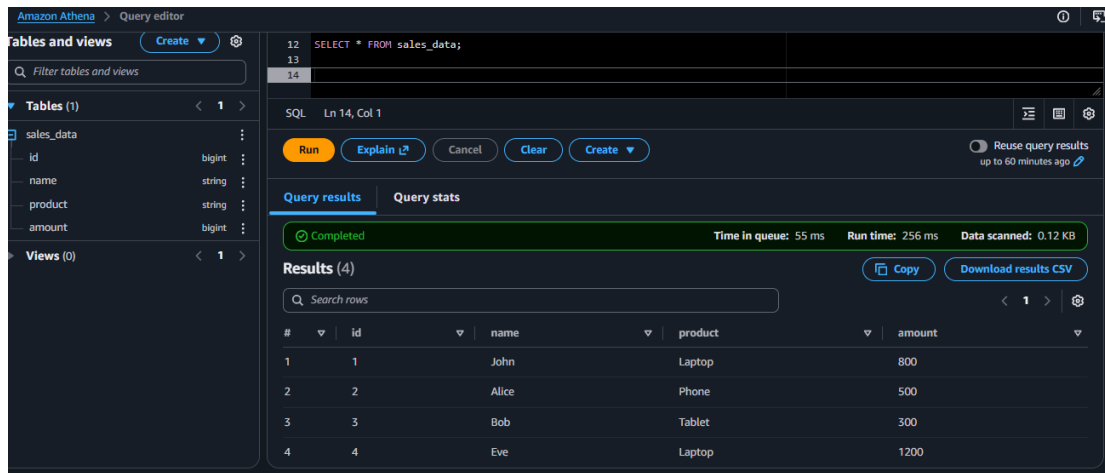
```
SELECT * FROM sales_sales_data;
```

Filter data

```
SELECT * FROM sales_sales_data WHERE amount > 500;
```

Aggregation

```
SELECT product, SUM(amount) FROM sales_sales_data GROUP BY product;
```



Key Concepts Demonstrated

- Data stored in S3 (not database)
- Glue creates metadata, not actual data
- Athena queries data directly from S3
- Multiple files behave as one dataset
- Fully serverless architecture

Troubleshooting

Issue 1: Only one file data visible

Cause: Crawler pointed to single file

Solution:

- Use folder path instead of file
- Re-run crawler

Issue 2: Athena query failed

Cause: Query result location not set

Solution:

- Configure S3 path for results

Issue 4: Missing data

Cause: New files added after crawler run

Solution:

- Re-run crawler

Conclusion

This project successfully demonstrates the implementation of a **serverless data lake architecture using AWS**. Data is stored in Amazon S3, cataloged using AWS Glue, and queried using Amazon Athena without provisioning any infrastructure. The system efficiently

handles multiple data files as a single dataset and allows scalable, cost-effective analytics. This architecture reflects real-world big data solutions and highlights key concepts such as data cataloging, serverless querying, and cloud-based storage.